Tutorial 3

- Partial Fraction
- Improper Integral → ① Just like a usual integral

  ② Direct Comparison

  ③ Limit Comparison

$x^{3/2} = x^{\frac{1}{2}} x^{\frac{2}{2}} = \sqrt{x} \cdot x$

Ex. 8.5

Q45 $\displaystyle\int \frac{1}{x^{3/2} - \sqrt{x}}\, dx = \int \frac{1}{\sqrt{x}(x-1)}\, dx$

$u = \sqrt{x}$

$du = \frac{1}{2\sqrt{x}}\, dx$

$\displaystyle = \int \frac{2}{u^2 - 1}\, du$

PF: $\frac{2}{u^2-1} = \frac{A}{u-1} + \frac{B}{u+1}$ ⇒ Solve for A, B

Q46

$$\int \frac{1}{(x^{1/3}-1)\sqrt{x}}\,dx$$

$\cancel{\text{Hint: } x = u^6}$

$u = \sqrt{x} \Rightarrow du = \frac{1}{2\sqrt{x}}\,dx$

$x = u^2$

$x^{1/3} = u^{2/3}$

$$= 2\int \frac{1}{u^{2/3}-1}\,du$$

$t = u^{1/3} \rightarrow u^{-2/3} = (u^{1/3})^{-2}$

$dt = \frac{1}{3}u^{-2/3}\,du$

$= \frac{1}{3}t^{-2}\,du$

$du = 3t^2\,dt$

$$= 2\int \frac{3t^2}{t^2-1}\,dt$$

$$= 6\int \frac{t^2}{t^2-1}\,dt$$

$\boxed{x^{m/n}} \leftarrow$

$u = x^{1/n}$

$du = \frac{1}{n}x^{-\frac{(n-1)}{n}}\,dx$

$= \frac{1}{n}(x^{\frac{1}{n}})^{-(n-1)}\,dx$

$= \frac{1}{n}u^{-(n-1)}\,dx$

↳ polynomial

$$\boxed{\frac{t^2-1+1}{t^2-1} = 1 + \frac{1}{t^2-1}}$$

$$= 6\int 1\,dt + 6\int \frac{1}{t^2-1}\,dt$$

↳ $(t-1)(t+1)$ → Use P.F.

Q50

$$\int \frac{1}{x^6(x^5+4)}\,dx \cdot \frac{x^4}{x^4}$$

$$= \int \frac{x^4}{\underbrace{x^{10}}_{u^2}(\underbrace{x^5}_{u}+4)}\,dx \qquad u = x^5$$

$$= \frac{1}{5}\int \frac{1}{u^2(u+4)}\,du$$

$$\frac{1}{u^2(u+4)} = \frac{A}{u} + \frac{B}{u^2} + \frac{C}{u+4}$$

$$\frac{1}{x^5(x^3+4)} \quad \times$$

$$\frac{1}{x^4(x^3+4)} \quad \checkmark$$

$$\frac{1}{x^m(x^n+c)}$$

$m+n-1$ is a multiple of n.

$$\frac{1}{x(x^4+1)}$$

Q54

$$\int \frac{\sqrt{x}}{\underbrace{\sqrt{2-\sqrt{x}}}_{u} + \underbrace{\sqrt{x}}_{\text{poly}}}\, \underbrace{dx}_{\text{poly}}$$

$\sqrt{x} \to$ poly

$u = \sqrt{2-\sqrt{x}}$

$u^2 = 2 - \sqrt{x}$

$\sqrt{x} = 2 - u^2$ (poly. in u)

$x = (2-u^2)^2$

Easy to differentiate

$dx = (\text{Poly. in } u)\, du$

Ex. 8.8

Q48 $\displaystyle\int_0^2 \frac{dx}{\underline{1-x}} = \int_0^1 \frac{dx}{1-x} + \int_1^2 \frac{dx}{1-x}$

$x=1$ is undefined

(a singular point)

$$\int_0^1 \frac{dx}{1-x} = \lim_{c\to 1^-} \int_0^c \frac{dx}{1-x} = \lim_{c\to 1^-} \Big[-\ln|1-x|\Big]_0^c$$

$$= \lim_{c\to 1^-} \Big[-\ln|1-c|\Big]$$

$$= \infty$$

Since $\int_0^1 \frac{dx}{1-x}$ diverges, the improper integral $\int_0^2 \frac{dx}{1-x}$ diverges.

Q56 $\displaystyle\int_2^\infty \frac{dx}{\sqrt{x^2-1}}$

~~Asymptotic~~

Asymptotically equivalence

$$\sqrt{x^2-1} \sim \sqrt{x^2} = \underline{\underline{x}}$$

Let $f(x) = \dfrac{1}{\sqrt{x^2-1}}$, $g(x) = \dfrac{1}{x}$.

$$\lim_{x\to\infty} \frac{f(x)}{g(x)} = \lim_{x\to\infty} \frac{\frac{1}{\sqrt{x^2-1}}}{\frac{1}{x}} = \lim_{x\to\infty} \frac{x}{\sqrt{x^2-1}}$$

$$\underline{\underline{\int_a^\infty \frac{1}{x}\,dx}}$$

$$= \lim_{x\to\infty} \frac{1}{\sqrt{1-\frac{1}{x^2}}}$$

$$= 1 > 0$$

Since $\displaystyle\int_2^\infty \frac{1}{x}\,dx$ diverges, $\displaystyle\int_2^\infty \frac{dx}{\sqrt{x^2-1}}$ diverges by Limit Comparison Test.

Q60 $\displaystyle\int_{\pi}^{\infty} \frac{1+\sin x}{\boxed{x^2}}\,dx$

$$\frac{\frac{1+\sin x}{x^2}}{\frac{1}{x^2}} = 1+\sin x$$

(LCT does not work)

Note that $1+\sin x \leq 2$ for $x \geq \pi$.

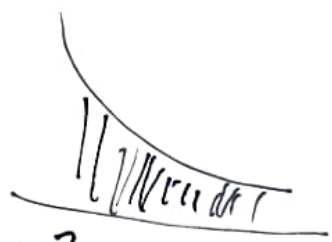
$$\Rightarrow \frac{1+\sin x}{x^2} \leq \frac{2}{x^2}$$

Since $\displaystyle\int_{\pi}^{\infty} \frac{2}{x^2}\,dx$ converges, $\displaystyle\int_{\pi}^{\infty} \frac{1+\sin x}{x^2}\,dx$ converges by Direct Comparison Test.

$$\int_{\pi}^{\infty} \frac{2}{x^2}\,dx = \cdots = \boxed{\frac{2}{\pi}}$$

$$\frac{2}{x^2} \to \frac{2x^{-1}}{-1} \to -2/x$$

$$\left[-2/x\right]_{\pi}^{\infty} = 0 - \left(-\frac{2}{\pi}\right) = \frac{2}{\pi}$$



Q64 $\int_e^\infty \ln(\ln x)\,dx$

$\ln x \leq \ln(\ln x)$ ~~#~~ $<$

$u = \ln x \Rightarrow x = e^u$

$du = \frac{1}{x}\,dx$

$dx = x\,du$

$= e^u\,du$

$x \to \infty,\ u \to \infty$

$x = e,\ u = \ln e = 1$

Correction!

$x = e^e$

So $u = \ln e^e$

$= e$

$= \int_{\boxed{1}}^\infty \ln(u) \cdot e^u\,du$

e

$\geq \int_{\boxed{1}}^\infty \ln(u)\,du$

e

Note that

$\ln(u) \leq \ln(u)\,e^u$ for $u \geq \boxed{1}$

e

~~Since~~ ~~for~~

$\frac{\ln(\ln x)}{\ln x} \to \frac{\frac{1}{x\ln x}}{\frac{1}{x}}$

$= \frac{1}{\ln x} \to 0$

(LCT does not work)

$$\int_{\boxed{1}\,e}^{\infty} \ln u \, du = \left[u \ln u - u \right]_{\boxed{1}\,e}^{\infty}$$

$$= \infty$$

$\boxed{\ln 1 = 0}$

$\ln 1.1 > 0$

Correction!

Note that

$$u(\ln u - 1) \geq \boxed{\ln u - 1} \quad \text{for } u \geq e$$

$$\Rightarrow \lim_{u \to \infty} u(\ln u - 1) = \infty$$

because $\lim_{u \to \infty} (\ln u - 1) = \infty$.

Alternative: Just note that

$$\lim_{u \to \infty} \underbrace{u}_{\infty} \underbrace{(\ln u - 1)}_{\infty} = \infty.$$